Sum of Four Values

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1642 | Sorting and Searching | https://cses.fi/problemset/task/1642

You are given an array of n integers, and your task is to find four values (at distinct positions) whose sum is x.

Input

The first input line has two integers n and x: the array size and the target sum.

The second line has n integers $a_1,a_2,\ldots,a_n$: the array values.

Output

Print four integers: the positions of the values. If there are several solutions, you may print any of them. If there are no solutions, print `IMPOSSIBLE`.

Constraints

- $1 \le n \le 1000$
- $1 \le x,a_i \le 10^9$

Example

Input
8 15 3 2 5 8 1 3 2 3

Output
2 4 6 7